University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

Total Marks:110

Final Term Exam Spring 2024

CSNC2413 – Computer Communications and Networks

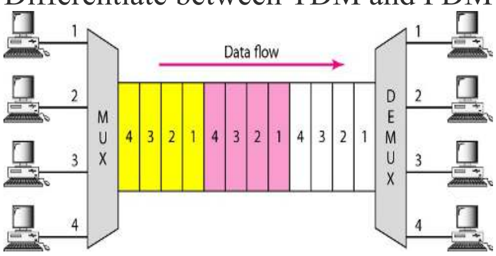
Duration: 2.5 Hours	Course Instructor:	Date/Time:
Roll No:	**Student Name:**	**SECTION:**

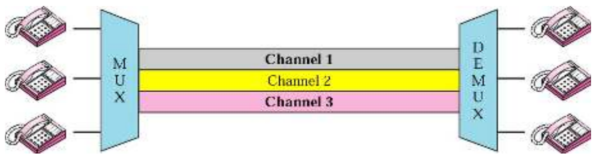
Follow the below-given instructions

1. Avoid cutting and over-writing.
2. Clearly mention the question number, **5 Bonus** marks for following order.
3. Answer all questions in the context of class discussions, handouts, and the text/reference book.
4. **Please provide answers to all questions on a separate answer sheet. Understanding the question is a part of exam.**

1-Short Questions Marks:4x5=20

a) Differentiate between TDM and FDM with help of Diagram.





b) For the given Ip's calculate the network address and broadcast address 100.87.32.15/11, 140.63.29.42/22.

   a. **100.64.0.0 /11 and 100.95.255.255 /11**
   b. **140.63.28.0 /22. And 143.63.31.255 / 22**

c) Compare Link state Algorithms with Distance vector algorithms and also mention 2 protocols for each.

Global or Decentralized…

global:

- all routers have complete topology, link cost info
- send link state to all nodes

  "link state" algo

OSPF, IS-IS

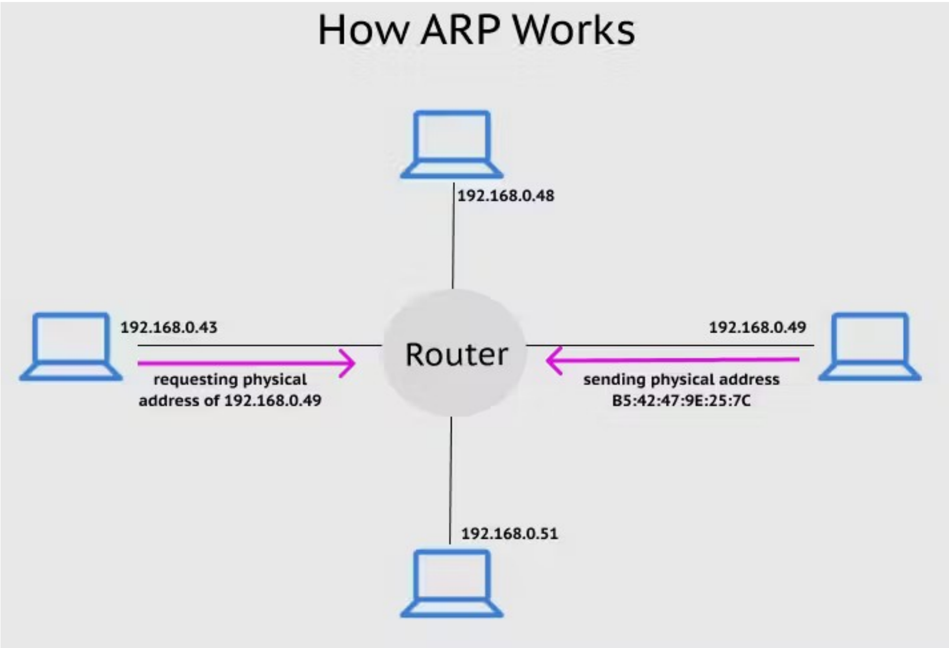
decentralized:

- router knows about physically connected neighbors *(link costs to neighbors)*
- exchange info with neighbors, perform iterative computations
  "distance vector" algo

RIP, IGRP

d) Describe the process how mac-addresses are learned by PC's in a local area network.

Using ARP



2-Long Questions Marks 15x6=90

A. Fill the table on answer sheet

Layer Names	**Data called**	**Protocols used (minimum 2 for each layer)**	**Addresses**
Application	PDU/ Message	DNS, HTTP, HTTPS	Application Specific Address, URLs
Transport	Segment	TCP, UDP	Port Address
Network	Datagram/ Packet	IP, IPX, DHCP	IP
Data Link	Frame	Ethernet, Aloha, CSMA	MAC
Physical	Bits/ Signals	N/A	N/A

B. As a network administrator you have been asked by your organization to develop a networking scheme. Choose a network address of your choice from class B **private Ip** address range and make subnets with following requirements.

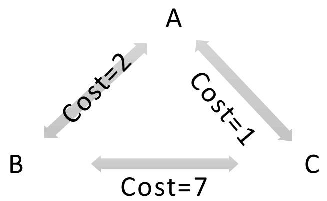
i. 1 subnet for 1000 hosts.

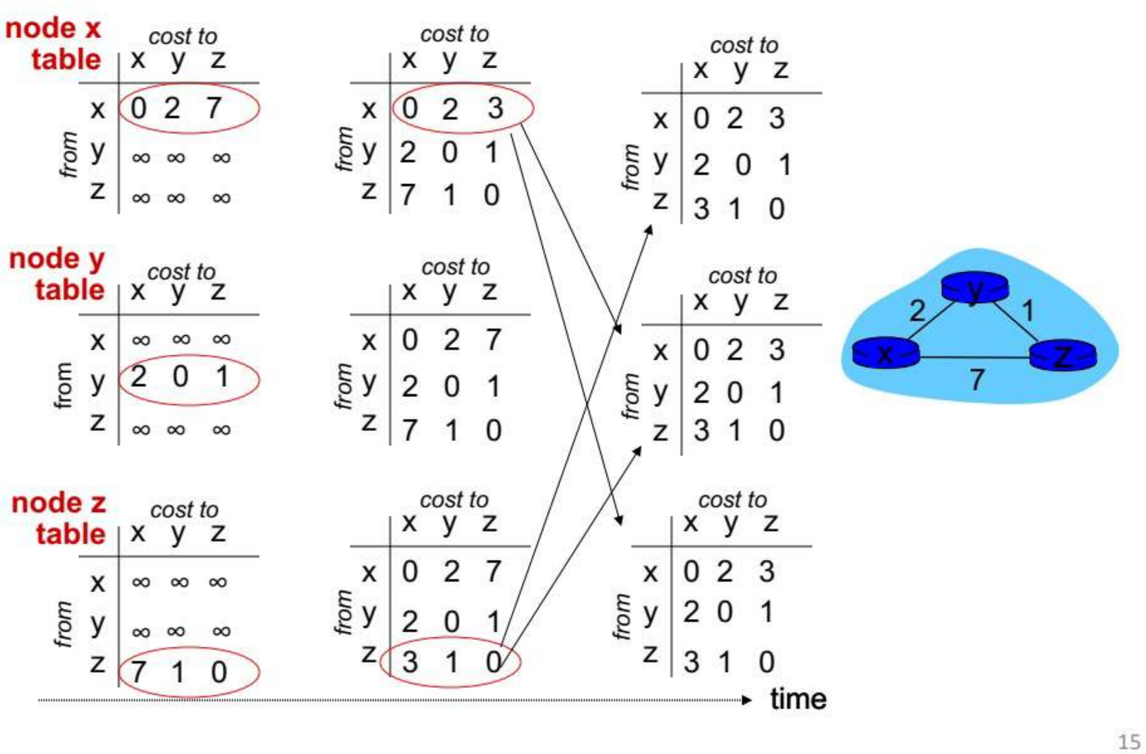
ii. 2 subnets for 2048 hosts.

iii. 3 subnets for 500 hosts and 2 subnets for 16 hosts each.

iv. For each subnet, you are required to write subnet mask, network address, broadcast address, and useful host addresses range.

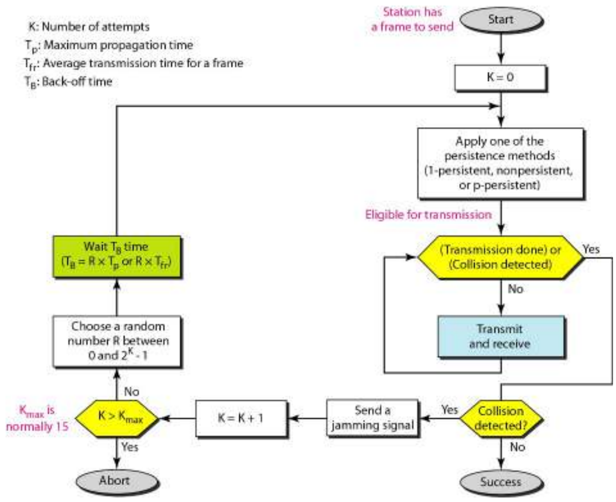
C. For given network show how routing updates are shared using **Bellman Ford equation.** You are required to show working using timing diagram as well. Once all the nodes are updated, consider link between B to C is upgraded and new cost is now 2, show how the updates will be shared again.

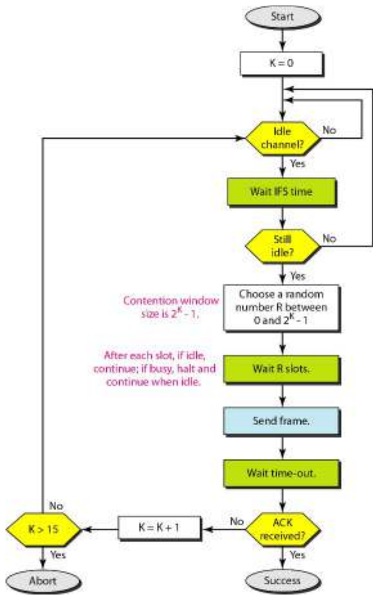




Same process to follow when the link gets updated.

D. Show the working of CSMA/CD and CSMA/CA using flow diagram





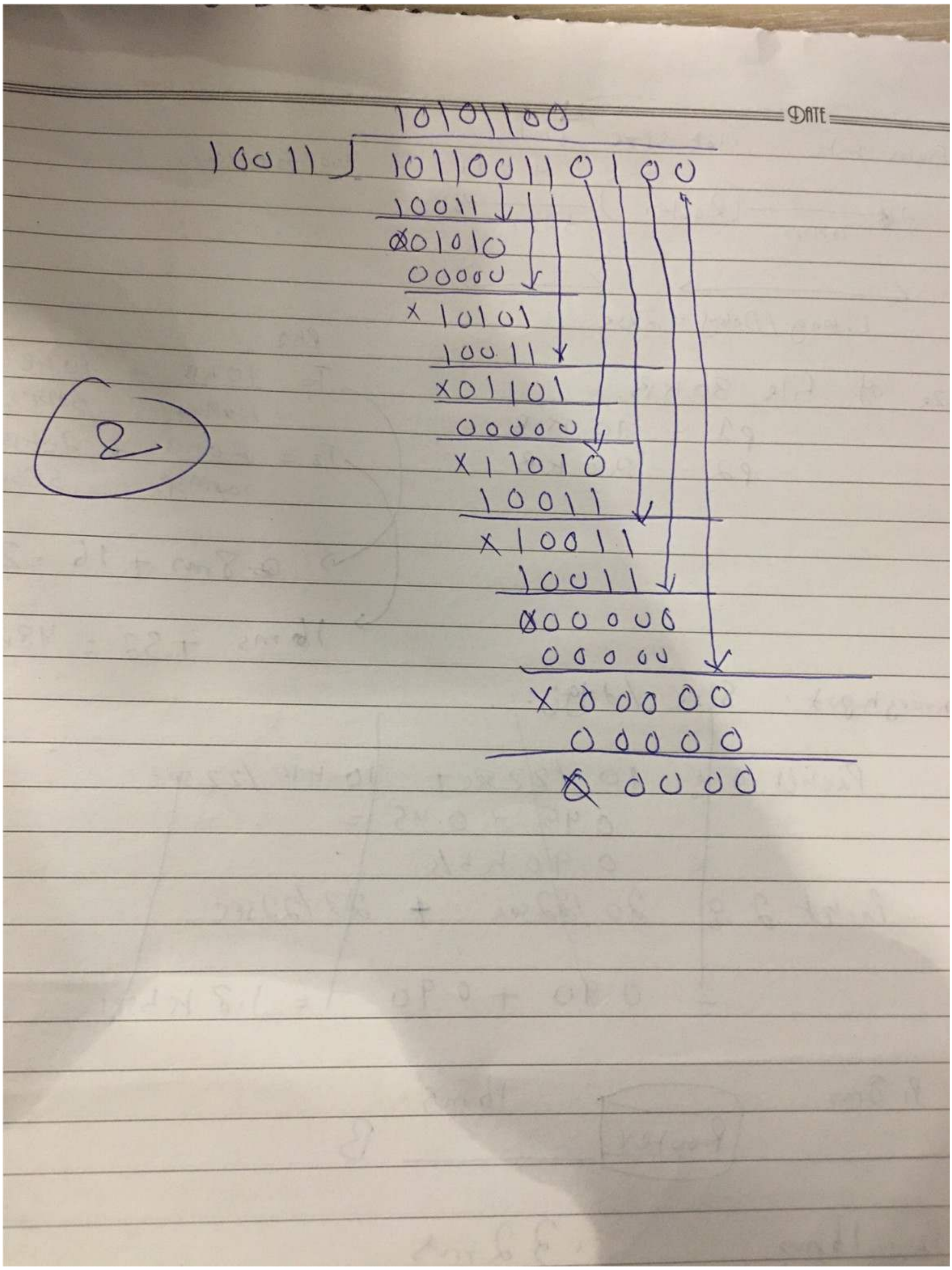
E. A bit stream 10110011 is to be transmitted using the standard CRC method. If generator polynomial/key is $x^4 + x + 1$ then show the working on both sender/receiver sides. Give Conclusion!

x^4+x+1

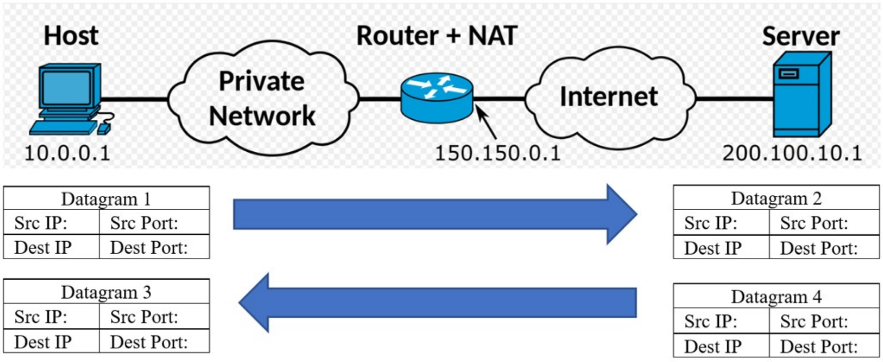
```
         10101100
10011 ) 101100110000
        10011
        x01010
         00000
        x10101
          10011
         x01101
          00000
          x1101
           11010
           10011
          x10010
            10011
           x00010
             00000
             x0010
```

```
00100
00000
x0100
```

1



F. Consider the figure given below. Suppose that the host with IP address 10.0.0.1 sends an IP datagram destined to server 200.100.10.1. The host uses source port as 6000, and the destination port as 80. Write the source and destination IP addresses, and source and destination port for each datagram numbered 1,2,3 & 4. **Assume NAT dynamically generated freely available port is 6001.**



Datagram 1

Datagram 1 : S 10.0.0.1 : 6000, D 200.100.10.1 :80

Datagram 2 S 200.100.10.1 : 6001, D 200.100.10.1 :80

Datagram 3 D 200.100.10.1 : 6001, S 200.100.10.1 :80

Datagram 4 D 10.0.0.1 : 6000, S 200.100.10.1 :80